PERSONAL INFORMATION

Muhammad Awais, PhD (Associate Professor)

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I have over 10 years of distinguished experience in research and teaching, and I am highly accomplished professional in the domain of Artificial Intelligence enabled digital health, smart wearable systems and autonomous systems. In 2018, Royal Academy of Engineering recognised my potential and endorsed my exceptional talent. I have authored over 60 research articles in well reputed and internally recognized journals and conferences, with h -index of 23, an i-10 index of 40, and over 2000 citations, demonstrating the impact and relevance of my work in the academic and research community. Being a Fellow of the Higher Education Academy (FHEA), I posses a comprehensive understanding of teaching pedagogy and a strong dedication to promoting educational excellence. In my various academic and leadership roles (Line manager for research staff, Engagement officer, a member of the Teaching and Learning Committee), I actively contribute to shaping academic frameworks and mentoring future professionals in my domain. I am a dedicated advocate for Equality, Diversity, and Inclusion (EDI), leading initiatives within my institution and beyond. As evidence of this commitment, I have served as a lead co-host for the In2STEM and In2Science programs, actively supporting their EDI efforts to provide opportunities for less privileged students.I am fellow of higher education academy (FHEA). Over the years, I have secured research grants, successfully delivered multiple projects, and supervised undergraduate, master's, and PhD dissertations. I have a proven track record of meeting deadlines, effectively managing teams, and working both independently and collaboratively. I am ambitious, embrace challenges, and excel in solving complex problems. Additionally, I have strong verbal and written communication skills and am proficient in presenting. Throughout my career, I have also developed and implemented new modules, assessments, training materials, and skill boot camps within higher education. I currently serve as an external examiner for higher education institutions (HEIs) in the UK, demonstrating my solid understanding of QAA standards.

Education

(2021 – 2022) **Postgraduate Certificate of Teaching in Higher Education (PGCTHE)**
Centre for Learning and Teaching, Edge Hill University, United Kingdom

(2014 – 2018) **PhD in Biomedical Electrical and System Engineering (IBES)**
Department of Electrical, Electronic, and Information Engineering "Guglielmo Marconi", University of Bologna, Italy (Awarded Fully Funded Scholarship from University of Bologna).

(2011 – 2014) **Master of Science (MS) in Electrical and Electronic Engineering**
Master by Research, Department of Electrical and Electronic Engineering, Universiti Teknologi, PETRONAS (UTP), Malaysia (Awarded Fully Funded Scholarship from UTP).

(2006 – 2010) **Bachelor of Science (BS) in Electronic Engineering**
Department of Electronic Engineering, Mohammad Ali Jinnah University (MAJU), Pakistan (Chancellor Scholarship granted due to high GPA 3.75/4.0).

Work Experience

July 2023 – Present **Associate Professor in Artificial Intelligence and Machine Learning**

Department of Computing Sciences, University of East Anglia, United Kingdom

- Working as a PI on two UKRI funded projects to develop AI enabled Cyber Security solution for IoMT to enhance their trustworthiness (please see further details in research fundings section)
- Accelerating research in data science and AI enabled cyber security to secure digital health devices and smart cities.
- Developing AI enabled digital health solution for wearable medical sensing devices.
- Acting as engagement officer at the department to improve the student retention, engagement, and productivity using variety of metrics and tools such as Tableau, BB, etc.
- Active research member of Cyber Security and data science research groups at UEA

Teaching Courses (undergraduate/postgraduate)

- Information Visualisation (Data Analytics, Data Visualisation, Power BI)
- Information Retrieval (Information extraction, data handling, processing)
- Ubiquitous computing (Applied AI, wearable sensing)

(July 2020 – July 2023 date)

Senior Lecturer in Artificial Intelligence and Data Analytics

Department of Computer Science, Faculty of Arts and Science, Edge Hill University, United Kingdom

- Worked on several internal and external data science, data analytics and AI based projects in the domain of digital health, industry 4.0, cloud computing and biomedical signal processing computing.
- Supervised Undergraduate, Master and Doctoral students in their dissertations.
- Co leader of Smart Systems and Applications Research (SSAR) research group.

Teaching Courses (undergraduate/postgraduate)

- Intelligent Systems (AI, Machine learning, Deep learning)
- Big Data and Visualisation (Teaching Data Analytics, Data Visualisation)
- Data Analytics (Data handling, processing and implementing AI algorithms)
- Project Management
- Programming Concepts to Construction (in Python & Java)

(Nov. 2019 – July 2020)

Research Fellow in Artificial Intelligence and Data Analytics

Faculty of Science and Engineering, University of Hull, United Kingdom

- Developed data analytics, artificial intelligence and machine learning based decision support systems and flood warning system on hydrology datasets, obtained through multi-modal sensing modalities (temporal, spatio-temporal). The developed AI and IoT based protypes and solutions will empower local SMEs to increase flood resilience and flood mitigation capabilities of UK/Europe.
- Developed AI driven cognitive decision support systems for Internet of Things deployed in industry and healthcare and build data analytics routines.

(Sep. 2018 – Oct. 2019)

Research Fellow in Data Analytics and Machine Learning enabled Digital Health

Centre for Immersive Technologies, University of Leeds, United Kingdom

- Developed a data analytics pipelines using Electroencephalogram (EEG) based close loop brain computer interface system through biomedical signal processing and machine learning to promote deep sleep, memory consolidation and learning through acoustic stimulation. This prototype detects sleep stages and improves sleep using AI, data analytics and acoustic stimulation and will be commercially available in 2021.
- Involved in various other data science projects i.e. analysing cycling behaviour in immersive and non-immersive environment settings using behavioural measures and EEG, accessing dental surgeons' performance in easy and difficult tasks using EEG and other behavioural parameters.
- Developed deep learning-based data mining approaches on time series physiological signals.

(Nov. 2017 – Aug. 2018)

Research associate in AI and digital health

Personal Health Systems Lab, Department of Electrical, Electronic, and Information Engineering "Guglielmo Marconi", University of Bologna, Italy

- Developed AI enabled ICT based wearable sensor systems to objectively quantify the human activity patterns in elderly population through data processing and data analytics. Several classical machine-learning and deep-learning based data processing methods were developed. These assisted technologies based novel development monitored elderly in their residential settings and generated activity profile that have help healthcare staff to make informed decisions.
- Worked on the European PreventIT project to predict the unlabelled activities of daily living performed by the elderly population through AI and data analytics. The motivation behind this approach was that the unlabelled data is much easier to accumulate than the labelled data since it only requires a data collection device (smartphone/smartwatch/body-worn sensor) carried by the subject.

(Nov. 2014 – Oct. 2017)

Graduate Scholar in Biomedical Signal Processing and Machine Learning

Personal Health Systems Lab, Department of Electrical, Electronic, and Information Engineering "Guglielmo Marconi", University of Bologna, Italy

- Developed wearable inertial sensors based physical activity classification system through data pre-processing, data analytics and AI to profile activities of daily living of elderly population in real life conditions. The data analysis utilized signal processing techniques, data mining approaches and machine learning algorithms to objectively quantity the underlying patterns from wearable sensors' data.

- Worked on the pilot study to annotate/label the activities of daily living using a wearable camera-based image processing system. The aim was to provide the ground truth information automatically to annotate the activities of daily living instead of manually marking the video recordings offline.

(Sep. 2015 – Nov. 2015)

Visiting Researcher in Biomechanics, AI and Machine Learning

Research Group on Geriatrics, Movement and Stroke, Department of Neuroscience, Faculty of Medicine, Norwegian University of Science and Technology (NTNU), Norway.

- Worked on the Norwegian ADAPT project (A Personalized Fall Risk Assessment System for promoting independent living) and implemented various state of the art AI and data analysis approaches for physical activity classification on elderly population to observe the weaknesses and strengths of existing systems in real life conditions using variety of signal processing and machine learning approaches.

(Jan. 2012 – Sep. 2014)

Research Assistant in Data Analytics and Biomedical Signal Processing

HICOE (Centre for Intelligent Signal & Imaging Research, CISIR), Department of Electrical and Electronic Engineering, Universiti Teknologi PETRONAS (UTP), Malaysia.

- Developed a novel data analytics approach for driver drowsiness detection by analysing the time data of physiological wearable sensors i.e. Electroencephalogram (EEG) sensors and Electrocardiogram (ECG) sensor. The data analysis is accomplished through biomedical signal processing, statistical analysis and implementation of machine learning approaches. The drivers drowsiness detection testbed was developed in collaboration with industrial partner i.e., Malaysian Institute of Road Safety and Research (MIROS).

Tutor of University Courses

- AI and Machine Learning
- Signal and Systems
- Digital/Biomedical Signal Processing
- Probability and Random Signals
- Structural Programming and Interfacing
- AI and Machine Learning

(Sep. 2010- Nov. 2011)

Junior Lecturer in Engineering

Department of Electronic Engineering, Mohammad Ali Jinnah University (MAJU), Pakistan.

- Worked as junior lecturer to assist students in experimentation of several courses apart from developing and devising course modules.
- Worked on the research project: Comparative analysis of energy aware routing protocols (GRACE-GRAdient Cost Establishment & GRAB (Gradient Broadcasting) in Java using sunspot wireless sensor networks.

Lab Teaching

- Digital Signal Processing
- Signal and Systems
- Physics
- Circuit Analysis
- Electronics

Research Fundings

Ongoing Projects

1. [Project lead] Proof of concept development of AI enabled Cyber Security solution for IoMT, Funding body: UKRI, £58,878, [PI, Successful], 2024-2025.
2. [Project lead] AI enabled multi-tier trust-management in Internet of Medical Things, Funding body: UKRI, £31,878, [PI, Successful], 2024.

Previous Attempts

1. Artificial Intelligence Driven Vulnerability Analysis and Proactive Measures for Risk Reduction in Covid Patients, Funding body: UKRI, £295k [Co-PI, unsuccessful].
2. Artificial Intelligence Driven Vulnerability Analysis and Proactive Measures for Risk Reduction in Covid Patients (AI-VAC), Funding body: RIF Edge Hill, £10k [Co-PI, unsuccessful].

3. Going Global Partnership PAK-UK Education Gateway Mobility Partnership for Faculty, Funding body: British council, £80k [Co-PI, unsuccessful].
4. A novel machine-learning based analytical framework for classifying children's physical activity, Funding body: RIF Edge Hill, £14k [PI, unsuccessful].

Trainings Delivered

Skills Bootcamp: Cloud Computing and Data Analytics (Two runs in year 2022)

- Skills Bootcamp organised and funded by UK department for Education (DfE), Skills for life and Institute of Coding (IoC), UK.
- Actively participated in the inception and development of Skills bootcamp for Cloud Computing and Data Analytics at Edge Hill University.
- Designed, developed and delivered the following short courses including their assessments:
  - Introduction to Artificial Intelligence and Machine Learning
  - Beginner's Python
  - Advanced Python
  - Advanced Data Science

SAS Training (2021- 2022)

- Worked as part of specialist training team to deliver SAS training.
- Delivered data analytics, business intelligence and data management operation through SAS.
- Conducted assessments and exams at the end of trainings

Dissertation: supervision, external examiner

External examiner (UG/PG/PhD)

- External Examiner for MSc Robotics and Artificial Intelligence at Cardiff Metropolitan University (2022-present).
- External Examiner for BSc Artificial Intelligence & Data Science at University of Northampton (2021-2024).
- 12-02-2024: External Examiner for the PhD Dissertation of Hira Hameed at University of Glasgow, United Kingdom- Thesis: Contactless AI enabled Hybrid sensing for cognitive Impairment.
- 03-01-2025: External Examiner for the PhD Dissertation of Muhammad Inam ur Rehman at National University of Science and Technology (NUST), Pakistan- Thesis: Automatic detection system for the classification of Tachycardias using Intracardiac Electrograms.
- 23/06/2021: External examiner for the PhD dissertation of Ms Anum Tanveer Kiyani, Middlesex University, United Kingdom - Thesis: Continuous User Authentication Featuring Behavioural Biometrics.
- 27/08/2021: External examiner for the PhD dissertation of Mr Mustafa Jan, Pakistan Navel Engineering College (PNEC), Karachi, Pakistan - Thesis: Risk Assessment Modelling Using Pattern Recognition & Data Mining

Thesis supervision (completed/on-going)

- Three PhD theses (completed)
- Four MS thesis (completed)
- Over 10 undergrad theses (completed)
- Several on-going dissertations (PhD/MS/BS)

Research Fundings

Successful funding Applications

- [Project lead] Proof of concept development of AI enabled Cyber Security solution for IoMT, Funding body: UKRI, £58,878, [PI, Successful], 2024-2025.
- [Project lead] AI enabled multi-tier trust-management in Internet of Medical Things, Funding body: UKRI, £31,878, [PI, Successful], 2024.

Unsuccessful funding Applications

- AI enabled Brain Sensing Framework for Fatigue Detection in Working Individuals: Promoting Digital Health Tech in Healthcare Sector to Drive Malysia's 2030 Vision, Funding Body: British Council, £159.9k 2024 [PI, unsuccessful].
- Artificial Intelligence Driven Vulnerability Analysis and Proactive Measures for Risk Reduction in Covid Patients, Funding body: UKRI, £295k [Co-PI, unsuccessful].
- Artificial Intelligence Driven Vulnerability Analysis and Proactive Measures for Risk Reduction in Covid Patients (AI-VAC), Funding body: RIF Edge Hill, £10k [Co-PI, unsuccessful].
- Going Global Partnership PAK-UK Education Gateway Mobility Partnership for Faculty, Funding body: British council, £80k [Co-PI, unsuccessful].
- A novel machine-learning based analytical framework for classifying children's physical activity, Funding body: RIF Edge Hill, £14k [PI, unsuccessful].

External Academic Activities

Guest Editor in Special Issues

- MDPI Sustainability: Special Issue - Industrial Internet of Things (IIoTs) and Industry 4.0, 2021.
- MDPI Sensors: Special Issue - Digital Healthcare in Pandemics, 2021.

Invited talks

- Keynote Speaker: Virtual conference on disruptive technologies in maritime sector: Digital health and IoT (ICDTMS 2021), June 28, 2021.
- Invited Talk: 2021 China-UK Researcher Links On-line Virtual Workshop on Employing Information and Communication Technology for Environmental Protection and Pollution Monitoring (ICT4EVER'21), 6-7 May 2021.

Conference Program committees

- Technical program committee member: SMARTTECH 2020, IoP 2019, ScalCom 2019

Research Interests, Skills & Achievements

Career Interest

- Applied Artificial Intelligence
- Wearable sensing and computing
- AI enabled Digital Health
- Cyber Security
- Data analytics aspects Internet of things devices
- Machine Learning and Deep Learning applications in smart: health, cities and factories

Achievements

- Received full-time scholarship for PhD at university of Bologna, Italy.
- Received full-time scholarship for Master Studies at Universiti Teknologi Petronas (UTP), Malaysia.
- Received Bronze Medal in Master thesis project in 33rd Science and Engineering Design Exhibition (SEDEX 33), UTP, Malaysia, 2014.
- Got distinction in my BS (Electronic Engineering) degree and received bronze medal.
- Dean's Honor Roll (Awarded eight times for semester Gpa > 3.50)
- Received 2nd prize in IDEAS (MAJU IEEE Chapter) Best Project Competition.

Language

Mother tongue(s)	Urdu
English	Excellent Command/ Highly proficient in spoken and written English
Italian	Basic Level of speaking and reading

Summer Schools, Trainings and Workshops

24th August, 2023	Equality and Diversity in the Workplace, University of East Anglia, UK
24th August, 2023	Safeguarding and Prevent in the Workplace, University of East Anglia, UK
(12 Sep.- 16 Sep. 2016)	Ubicomp Doctoral School: Wearable Sensing, Health and Physical Activity, Heidelberg, Germany
(27 June- 30 June 2016)	Connected Health Summer School: Smarter Environments for Smarter Living, Florence, Italy.
(18 April - 20 April 2016)	European Spring School on "Electromagnetics and emerging technologies for pervasive applications: Internet of Things, Health and Safety", Marconi Foundation, Bologna, Italy.
(23 Feb.- 24 Feb. 2016)	The European Union (EU) Falls Festival: Implementation of Innovation into Policy and Practice, Bologna, Italy.
(25 August 2015)	Tutorial on Movement analysis via wearable inertial sensors: state of the art and perspectives, EMBC, Milan, Italy.
(24 Mar.- 25 Mar. 2015)	The European Union (EU) Falls Festival: Technology in the prediction, detection and prevention of falls, Robert-Bosch-Hospital, Stuttgart, Germany
(24 Sep.- 28 Sep. 2012)	Short course on Biostatistics for sample size calculation and data analysis, Universiti Teknologi PETRONAS (UTP), Malaysia.

Certifications & Memberships

March, 2018

Deep Learning Specialization

Organizer: coursera.org, deeplearning.ai
Tutor: Prof. Andrew Ng
Affiliation: Stanford University
Specializations:

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models (Recurrent Neural Networks -RNNs)

MEMBERSHIPS

- Fellow of Higher Education Academy (FHEA), UK
- Member, British Computer Society (BCS), London, United Kingdom
- Member, Institute of Electrical and Electronic Engineering (IEEE) - Membership # 92613895
- Member, IEEE Engineering in Medicine and Biology Society (EMBS) - Membership # 92613895
- Member, Pakistan Engineering Council (PEC) - Membership # ELECTRO/13713

ONLINE PRESENCE

Google scholar: https://scholar.google.com/citations?user=9TPgVvIAAAAJ&hl=en
Linkedin: https://www.linkedin.com/in/muhammad-awais-phd-a411aa6a/
Orcid: https://orcid.org/my-orcid?orcid=0000-0001-6421-9245

REF 2021 Contributions

[1] **M. Awais**, Chiari, L., Ihlen, E. A. F., Helbostad, J. L., & Palmerini, L. "Physical Activity Classification for Elderly People in Free-Living Conditions, In *IEEE journal of biomedical and health informatics (JBHI)*, 23(1), 197-207, 2019.

[2] M. Raza, **M. Awais**, W. Ellahi, H. X. Nguyen, N Aslam, H. Le-minh, "Diagnosis and monitoring of Alzheimer's patients using classical and deep learning techniques." in *Expert Systems with Applications*, Vol 136, pp. 353-364, 2019.

[3] M. Raza, **M. Awais**, N. Singh, M. Imran, S. Hussain, "Intelligent IoT Framework for Indoor Healthcare Monitoring of Parkinson's Disease Patient", In *IEEE Journal on Selected Areas in Communications*, 2020.

4 Significant Publications

[1] **M. Awais**, M. Raza, N. Singh, K. Bashir, U. Manzoor, S. Islam, J.JPC Rodrigues. "LSTM-Based Emotion Detection Using Physiological Signals: IoT Framework for Healthcare ansd Distance Learning in COVID-19." IEEE Internet of Things Journal 8, no. 23 (2020): 16863-16871.

[2] **M. Awais**, Chiari, L., Ihlen, E. A. F., Helbostad, J. L., & Palmerini, L. "Physical Activity Classification for Elderly People in Free-Living Conditions, In *IEEE journal of biomedical and health informatics (JBHI)*, 23(1), 197-207, 2019.

[3] M. Raza, **M. Awais**, W. Ellahi, H. X. Nguyen, N Aslam, H. Le-minh, "Diagnosis and monitoring of Alzheimer's patients using classical and deep learning techniques." in *Expert Systems with Applications*, Vol 136, pp. 353-364, 2019.

[4] M. Raza, **M. Awais**, N. Singh, M. Imran, S. Hussain, "Intelligent IoT Framework for Indoor Healthcare Monitoring of Parkinson's Disease Patient", In *IEEE Journal on Selected Areas in Communications*, 2020.

Selected Publications

Journals

[1] Shah, S. A. A., Shahzad, A., Alhussein, M., Goh, C. M., Aurangzeb, K., Tang, T. B., & **Awais, M.**, "An implementation of multiscale line detection and mathematical morphology for efficient and precise blood vessel segmentation in fundus images". Computers, Materials & Continua, 79(2), 2565-2583, 2024.

[2] Mushtaq, F., Welke, D., Gallagher, A., Pavlov, Y. G., Kouara, L., Bosch-Bayard, J., **M. Awais**,& Valdes-Sosa, P. (2024). One hundred years of EEG for brain and behaviour research. *Nature Human Behaviour*, 1-7.

[3] Ali Shah, S. A., Shahzad, A., Alhussein, M., Goh, C. M., Aurangzeb, K., Tang, T. B., **M. Awais** (2024). An Implementation of Multiscale Line Detection and Mathematical Morphology for Efficient and Precise Blood Vessel Segmentation in Fundus Images. Computers, Materials & Continua, 79(2).

[4] Alsareii, S. A., Raza, M., Alamri, A. M., AlAsmari, M. Y., Irfan, M., Raza, H., & **Awais, M**. (2023). Artificial Intelligence and Internet of Things Enabled Intelligent Framework for Active and Healthy Living. CMC, 75(2).

[5] Ahmad, J., **Awais, M**., Rashid, U., Ngamcharussrivichai, C., Naqvi, S. R., & Ali, I. (2023). A systematic and critical review on effective utilization of artificial intelligence for bio-diesel production techniques. Fuel, 338, 127379.

[6] Yaqoob, S., Hussain, A., Subhan, F., Pappalardo, G., & **Awais, M**. (2023). Deep Learning Based Anomaly Detection for Fog-Assisted IoVs Network. IEEE Access, 11, 19024-19038.

[7] Almasabi, Saleh, Turki Alsuwian, **M. Awais**, Muhammad Irfan, Mohammed Jalalah, Belqasem Aljafari, and Farid A. Harraz. "False Data Injection Detection for Phasor Measurement Units." Sensors 22, no. 9 (2022): 3146.

[8] Alsareii, S. A., **M. Awais,**, Alamri, A. M., AlAsmari, M. Y., Irfan, M., Aslam, N., & Raza, M. (2022). Physical Activity Monitoring and Classification Using Machine Learning Techniques. Life, 12(8), 1103.

[9] Alsareii, S. A., Raza, M., Alamri, A. M., AlAsmari, M. Y., Irfan, M., Khan, U., & **Awais, M**. (2022). Machine Learning and Internet of Things Enabled Monitoring of Post-Surgery Patients: A Pilot Study. Sensors, 22(4), 1420.

[10] S.A. Alsareii, M. Raza, A.M. Alamri, M.Y. AlAsmari, M. Irfan, U. Khan, **M. Awais**. "Machine Learning and Internet of Things Enabled Monitoring of Post-Surgery Patients: A Pilot Study." Sensors 22, no. 4 (2022): 1420.

[11] **M. Awais**, Chiari, L., Ihlen, E.A., Helbostad, J.L. and Palmerini, L., "Classical Machine Learning Versus Deep Learning for the Older Adults Free-Living Activity Classification". Sensors, 21(14), p.4669, 2021.

[12] **M. Awais**; M. Raza; N. Singh, U. Manzoor, S. Islam, "IoT enabled intelligent emotions evaluation framework to enhance distance learning in COVID-19", in *IEEE Internet of Things Journal*, 2021.

[13] Bogacz, M., Hess, S., Calastri, C., Choudhury, C. F., Mushtaq, F., **Awais, M**., ... & Erath, A. Modelling risk perception using a dynamic hybrid choice model and brain-imaging data: an application to virtual reality cycling. Transportation research part C: emerging technologies, 133, 103435, 2021.

[14] M. U. Hadi, **M. Awais**, M. Raza, K. Khurshid, H. Jung. "Neural Network DPD for Aggrandizing SM-VCSEL-SSMF-Based Radio over Fiber Link Performance." *In Photonics*, vol. 8, no. 1, p. 19, 2021.

[15] M.K. Ghori, **M. Awais**, A.S. Khattak, M. Imran, L. Szathmary. "Treating class imbalance in non-technical loss detection: An exploratory analysis of a real dataset." IEEE Access 9 (2021): 98928-98938.

[16] R.K. Matthew, F. Mushtaq, Immersive Healthcare Collaboration. "Three principles for the progress of immersive technologies in healthcare training & education." BMJ Simulation & Technology Enhanced Learning (2021).

[17] S. Yaqoob, A. Ullah, **M. Awais**, I. Katib, Ai. Albeshri, R. Mehmood, M. Raza, S. Islam, J. JPC Rodrigues.

"Novel congestion avoidance scheme for Internet of Drones." Computer Communications 169 (2021): 202-210.

[18] M. Irfan, A.S. Alwadie, A. Glowacz, M. Awais *et al*. "A Novel Feature Extraction and Fault Detection Technique for the Intelligent Fault Identification of Water Pump Bearings." *Sensors* 21, no. 12 (2021): 4225.

[19] P. D. E. Baniqued, E.C. Stanyer, **M. Awais**, A. Alazmani, A.E. Jackson, M. Mon-Williams, M. A., F. Mushtaq, R.J. "Brain-Computer Interface Robotics for Hand Rehabilitation After Stroke: A Systematic Review". *Journal of NeuroEngineering and Rehabilitation,* 18(1), 1-25, 2021.

[20] K. Ali, H.X. Nguyen, Q.T.V., P. Shah, M. Raza, V.V. Paranthaman, B. Rahmadi, **M. Awais**, S. Islam, J.J.P.C Rodrigues. "Review and Implementation of Resilient Public Safety Networks: 5G, IoT, and Emerging Technologies." IEEE network 35, no. 2 (2021): 18-25.

[21] **M. Awais**, M. Raza, N. Singh, K. Bashir, U. Manzoor, S. Islam, J.JPC Rodrigues. "LSTM-Based Emotion Detection Using Physiological Signals: IoT Framework for Healthcare ansd Distance Learning in COVID-19." IEEE Internet of Things Journal 8, no. 23 (2020): 16863-16871.

[22] M. Khalid, **M. Awais**, N. Singh, S. Khan, M. Raza, Q.B. Malik, M. Imran. "Autonomous transportation in emergency healthcare services: framework, challenges, and future work." IEEE Internet of Things Magazine 4, no. 1 (2021): 28-33.

[23] E.C. Stanyer, **M. Awais**, P. D. Baniqued, L. Kouara, A. Davies, T. Killan, F. Mushtaq, "The impact of acoustic stimulation during sleep on memory and sleep architecture: A meta - analysis." Journal of sleep research ,e13385, 2021.

[24] M. Raza, N. Singh, M. Khalid, S. Khan, **M. Awais**, M.U. Hadi, M. Imran, S. Islam, J.P.C Rodrigue, "Challenges and Limitations of Internet of Things Enabled Healthcare in COVID-19," in IEEE Internet of Things Magazine, vol. 4, no. 3, pp. 60-65, September 2021, doi: 10.1109/IOTM.0001.2000176.

[25] A Balkhoyor, **M. Awais**, F. Mushtaq "Performance analysis of dental surgeons using EEG based neuro imaging techniques and behavioral measures", in *BMJ Surgery, Interventions, & Health Technologies*, 2020.

[26] M. Raza, **M. Awais**, K. Ali, N. Aslam, V.V. Paranthaman, M. Imran, "Establishing effective communications in disaster affected areas and artificial intelligence-based detection using social media platform", in *Future Generation Computer Systems*, 112, 1057-1069, 2020.

[27] M. Raza, **M. Awais**, N. Singh, M. Imran, S. Hussain, "Intelligent IoT Framework for indoor healthcare monitoring of Parkinson's Disease Patients", In *IEEE Journal on Selected Areas in Communications*, 2020.

[28] M. Bogacz, S. Hess, C. Choudhury, C. Calastri, F. Mushtaq, **M. Awais**, A. Erath, M.V. Eggermond M. Nazemi,"Cycling in virtual reality: modelling behaviour in an immersive environment" *Transportation Letters: The International Journal of Transportation Research,* 2020, pp. 1-16.

[29] S. Rahman, M. Irfan, M. Raza, K.M. Ghori, S. Yaqoob, **M. Awais**, "Performance Analysis of Boosting Classifiers in Recognizing Activities of Daily Living.", in *International Journal of Environmental Research and Public Health*, 17(3), 1082, 2020.

[30] M. Bogacz, S. Hess, C. Choudhury, C. Calastri, F. Mushtaq, M.V. Eggermond, A. Erath, **M. Awais**, "A Comparison of Cycling Behaviour between Keyboard-Controlled and Instrumented Bicycle Experiments in Virtual Reality", *Transportation Research Record*, 2020.

[31] M. Raza, **M. Awais**, W. Ellahi, H. X. Nguyen, N Aslam, H. Le-minh, "Diagnosis and monitoring of Alzheimer's patients using classical machine learning and deep learning techniques." in *Expert Systems with Applications*, Vol 136, pp. 353-364, 2019.

[32] **M. Awais** et al. "An Internet of Things based bed-egress alerting paradigm using wearable sensors in elderly care environment." *Sensors*, Vol 11, 2498, 2019.

[33] K. M. Ghori, R.A. Abbasi, **M. Awais**, M. Imran, A. Ullah, L. Szathmary, "Performance Analysis of Different Types of Machine Learning Classifiers for Non-Technical Loss Detection." in *IEEE Access*, 8, 16033-16048, 2019.

[34] **M. Awais**, Chiari, L., Ihlen, E. A. F., Helbostad, J. L., & Palmerini, L. "Physical Activity Classification for Elderly People in Free-Living Conditions". In *IEEE journal of biomedical and health informatics (JBHI)*, 23(1), 197-207, 2019.

[35] M. Irfan, N. Saad, A. Alwadie, **M. Awais**, M.A. Sheikh, A. Glowacz, V. Kumar, "An Automated Feature Extraction Algorithm for Diagnosis of Gear Faults." In *Journal of Failure Analysis and Prevention*, 1-8, 2018.

[36] **M. Awais**, N. Badruddin, M. Drieberg, "A hybrid approach to detect driver drowsiness utilizing physiological signals to improve system performance and wearability." In *Sensors,* vol. 17, no. 9, pp 1991, 2017.

[37] **M. Awais**, L. Palmerini, A. Bourke, E.A. Ihlen, J. Helbostad, L. Chiari, "Performance Evaluation of State of the Art Systems for Physical Activity Classification of Older Subjects Using Inertial Sensors in a Real Life Scenario: A Benchmark Study", in *Sensors*, vol. 16, no. 12, pp. 2105, 2016.

Conferences

[38] Samwini, H., Awais, M., & Pereira, E. (2023, July). Middleware for resource sharing in fog computing with IoT applications. In IEEE EUROCON 2023-20th International Conference on Smart Technologies (pp. 508-513). IEEE.

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